

Final System Project Paper

SIA2

Lost & Found System

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Introduction

The Lost & Found System is a web-based application designed to help organizations, schools, and public facilities manage lost and found items efficiently. Many institutions still rely on manual logbooks or verbal reporting when handling lost items, which often results in misplaced records, unclaimed items, and difficulty in tracking ownership.

Because of these challenges, The system provides a centralized platform where users can submit reports, search for items, and manage claims in an organized and efficient way.

Background of the Study

In many schools and public places, lost and found processes are handled manually. Items are often recorded in notebooks or stored without proper documentation, making it difficult to track or return them to their rightful owners.

Manual systems are inefficient, prone to human error, and can lead to confusion or loss of important information about items. As the number of users and items increases, managing records becomes even more difficult.

To solve these issues, the Lost & Found System was developed. It digitizes the reporting, tracking, and claiming of lost and found items, improving accuracy, accessibility, and efficiency.

Purpose of the study/project

The main purpose of the system is to provide a fast and organized way of managing lost and found items. The system allows users to report lost or found items online and enables administrators to manage and verify item records.

It also helps users search for their lost belongings easily and claim items in a structured and secure process.

Importance of the project

This project is important because it improves the efficiency of lost and found management. It reduces manual paperwork, minimizes errors, and ensures better tracking of items.

It also improves communication between users and administrators, making the process of returning lost items faster and more reliable.

Problems the System Aims to Solve

The Lost & Found System addresses several problems commonly experienced in manual lost and found processes.

One major issue is the lack of proper documentation, which often leads to missing or incomplete records of lost items. Another problem is difficulty in tracking items, especially when multiple people report similar descriptions.

Manual systems also cause delays in returning items and can lead to confusion or misplaced belongings. Additionally, there is no centralized database in traditional systems, making it hard to manage large volumes of reports.

This system solves these problems by providing a digital and centralized platform for managing lost and found items efficiently.

Objectives of the System

General Objective

To develop a web-based Lost & Found System that manages item reporting, tracking, and claiming in an efficient and organized manner. **Specific Objectives**

1. To create a digital database for lost and found item reports.
2. To allow users to report lost or found items online.
3. To provide a search feature for locating items.
4. To improve tracking and organization of item records.
5. To implement secure user authentication.
6. To reduce manual paperwork and improve efficiency.

Scope and Limitations

Scope

The system is designed to:

- Allow users to report lost and found items
- Store item details and descriptions
- Allow searching of lost items
- Manage item claims and status updates

- Provide secure login for admin and users
- Maintain a centralized item database
- Improve tracking and retrieval of lost items
- Reduce manual record keeping

Limitations

The system is limited to:

- Web-based access only
- Registered users (admin and users)
- Text and image-based item reports only
- Small to medium-scale institutions

The system does not include:

- Mobile application version
- Payment or financial transactions
- Courier or delivery integration
- AI-based image recognition for items
- SMS or email notification system (optional future improvement)

Technologies Used

Programming Languages

- **HTML** – structure of web pages
- **CSS** – design and styling
- **JavaScript** – interactivity and dynamic features
- **PHP** – backend logic

Frameworks/Libraries

- **Laravel** – main backend framework
- **Bootstrap** – responsive UI design
- **JWT Authentication** – secure login system
- **Session Management** – user session handling

Database Management System

- **MySQL Database** – stores lost and found item data
- **phpMyAdmin** – database management tool

Development Tools Used

- **Visual Studio Code** – code editor
- **GitHub** – version control system
- **XAMPP** – local server environment

System Design

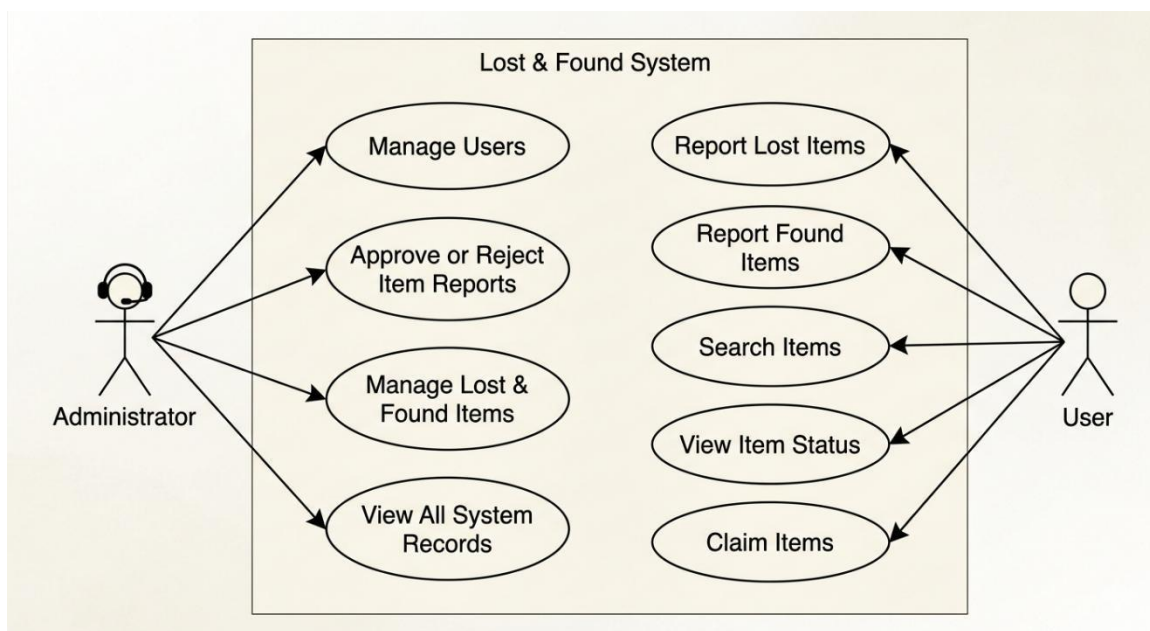
Use Case Diagram

Administrator

- Manage users
- Approve or reject item reports
- Manage lost and found items
- View all system records

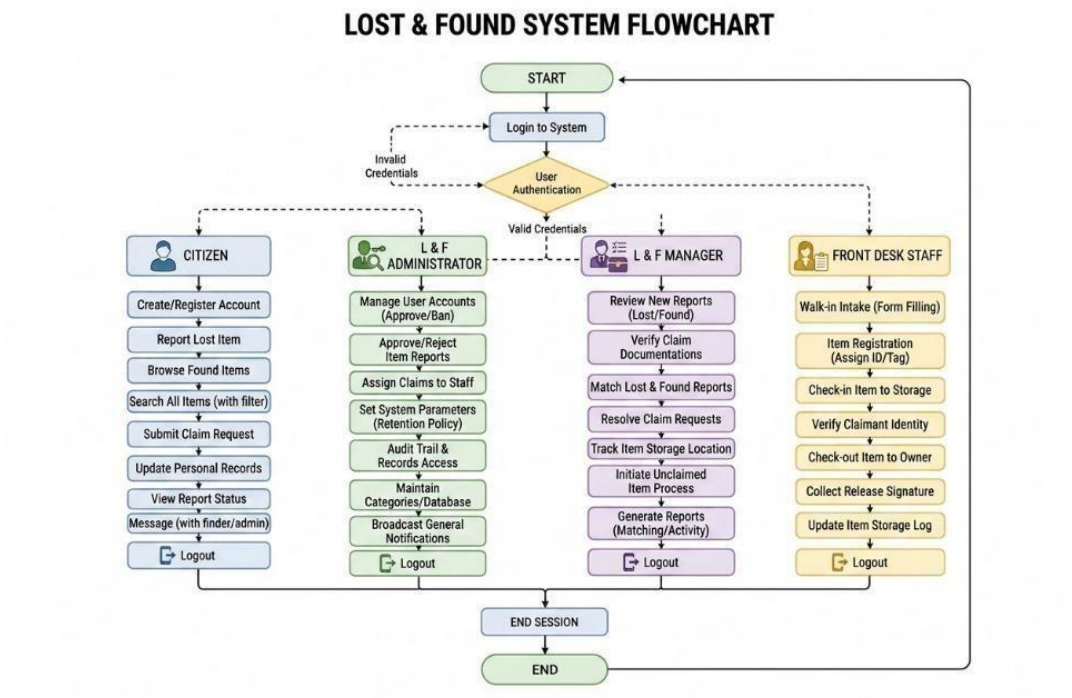
User

- Report lost items
- Report found items
- Search items
- View item status
- Claim items



Flowchart Description

The system begins when the user logs in or registers an account. After authentication, users are directed to their dashboard. Users can report lost or found items, search for items, or view item status. Administrators manage all reports, verify items, and update statuses. All data is stored in the database for processing and retrieval.

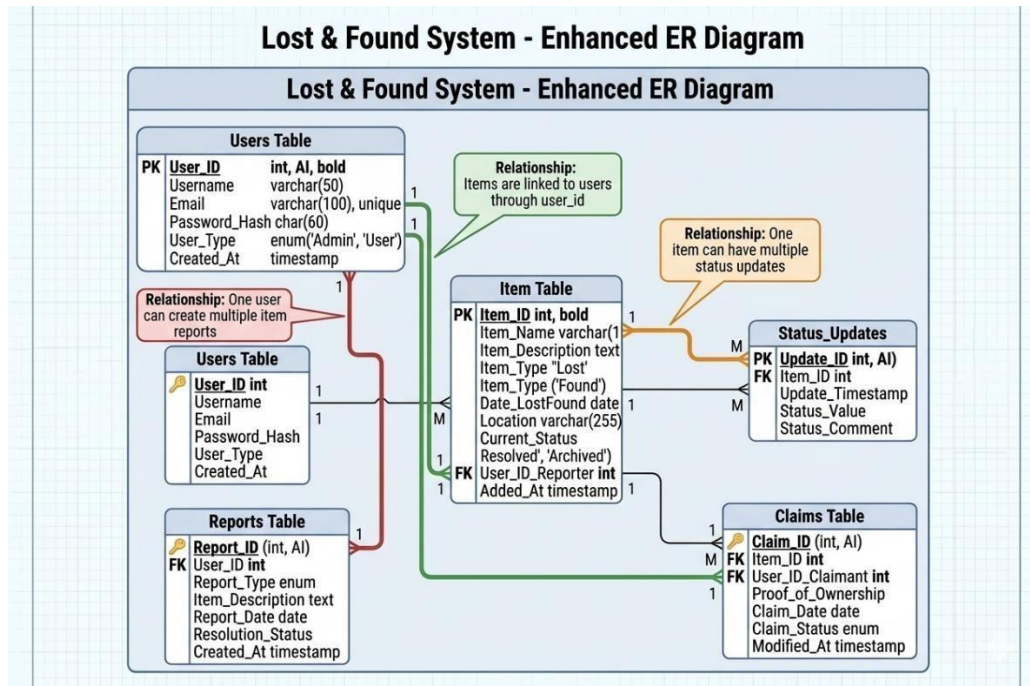


ER Diagram / Database Design

The database design consists of several related tables:

- **Users Table** – stores user accounts
- **Item Table** – stores lost and found item details
- **Reports Table** – stores item reports
- **Claims Table** – stores claim requests
- **Relationships**
- One user can create multiple item reports

- One item can have multiple status updates
- Items are linked to users through user_id



System Architecture Diagram

1. Client–Server Architecture

Client Side (Web Browser Interface) Used by: Admin and user Functions:

- Login and registration
- Item reporting
- Searching items
- Viewing results

2. Application Server (Laravel) Handles:

- Business logic
- Authentication
- Item management
- Routing and API requests

3. Database Server (MySQL)

Stores:

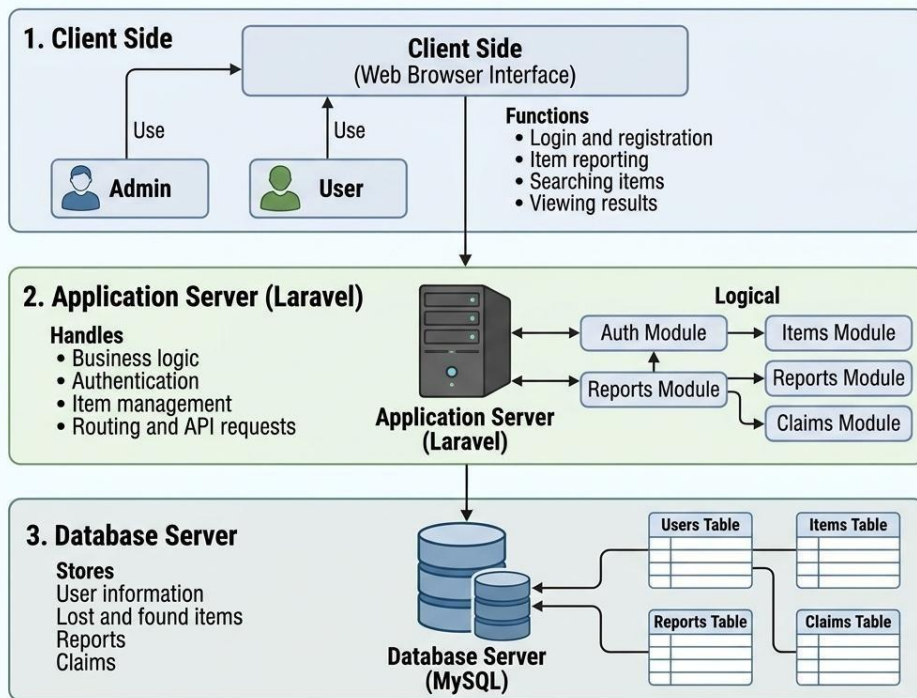
User information

Lost and found items

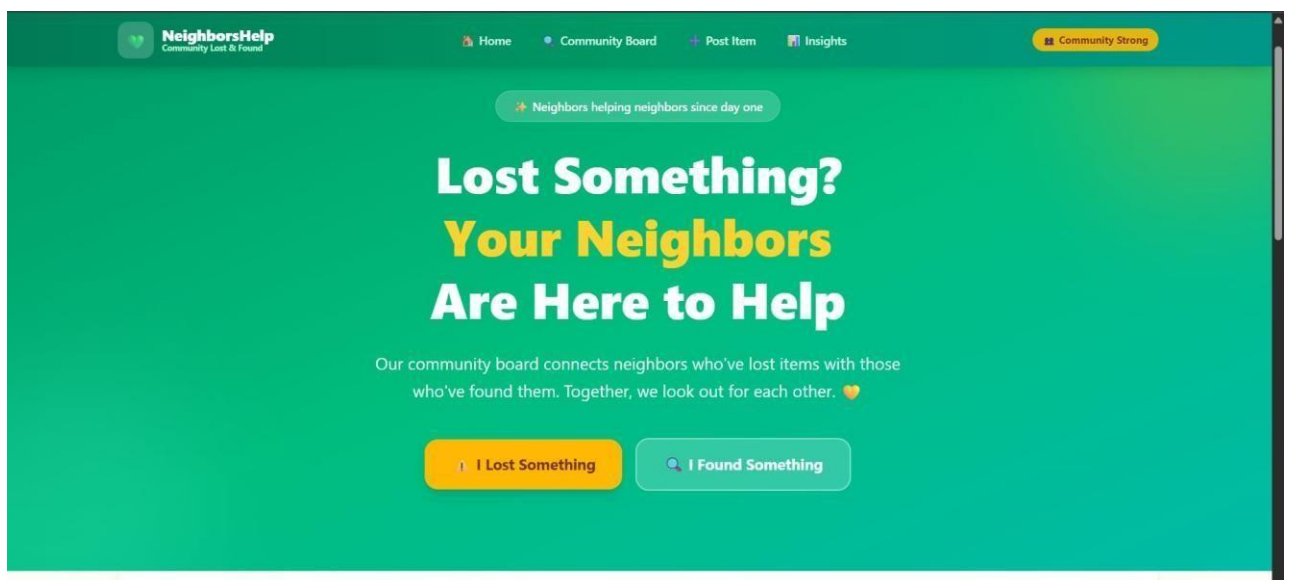
Report

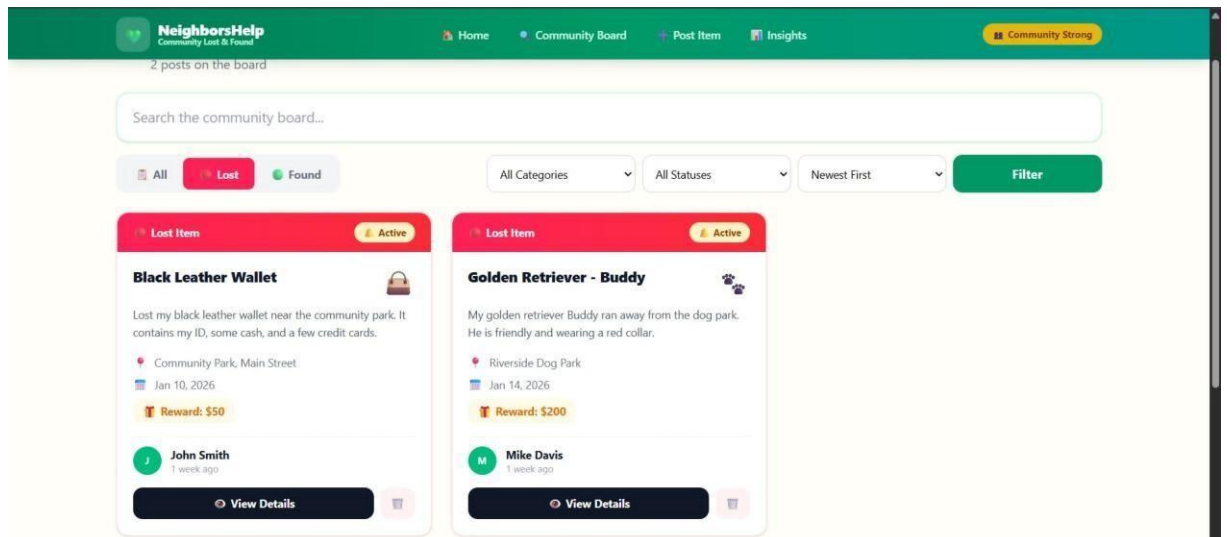
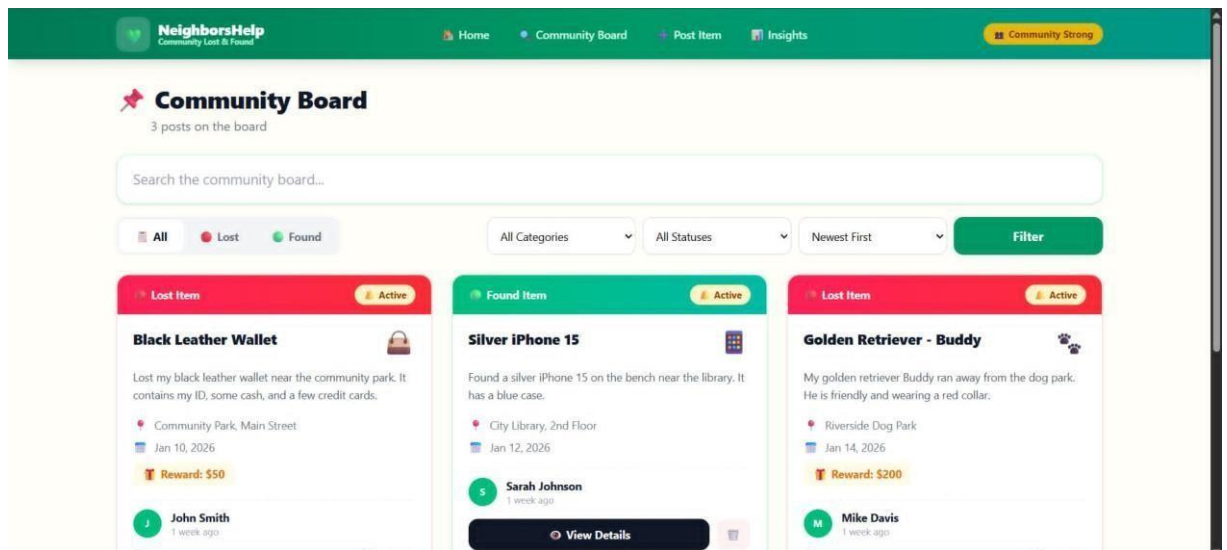
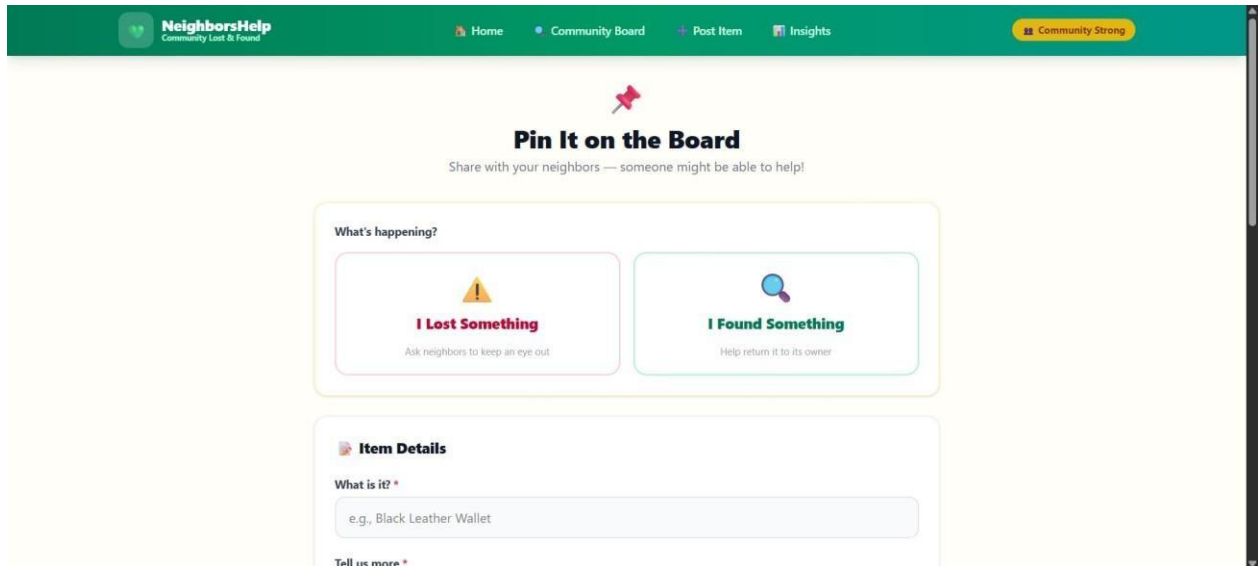
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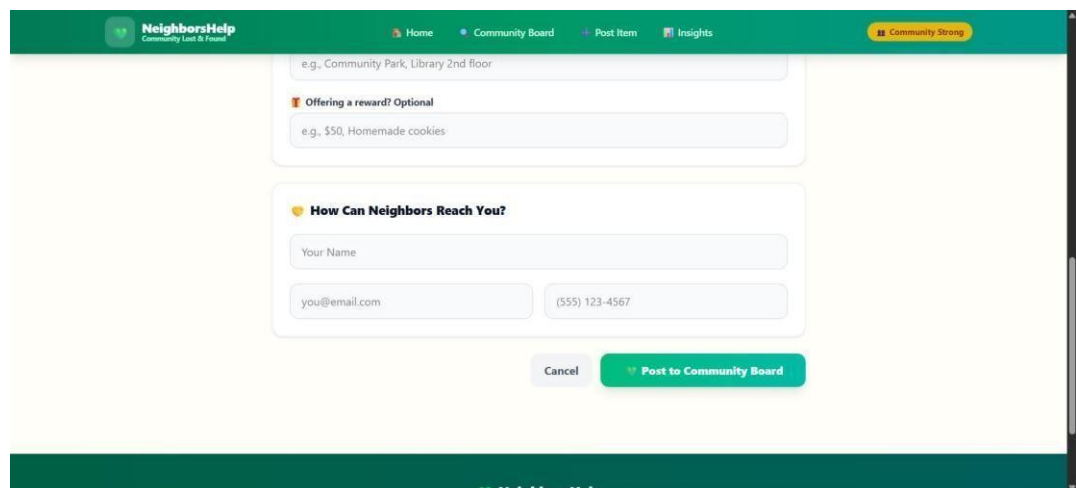
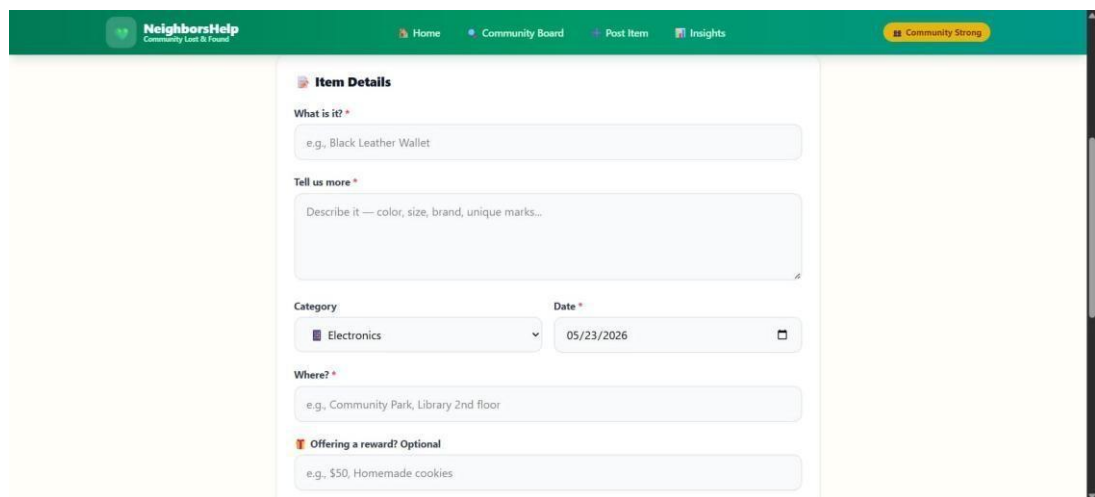
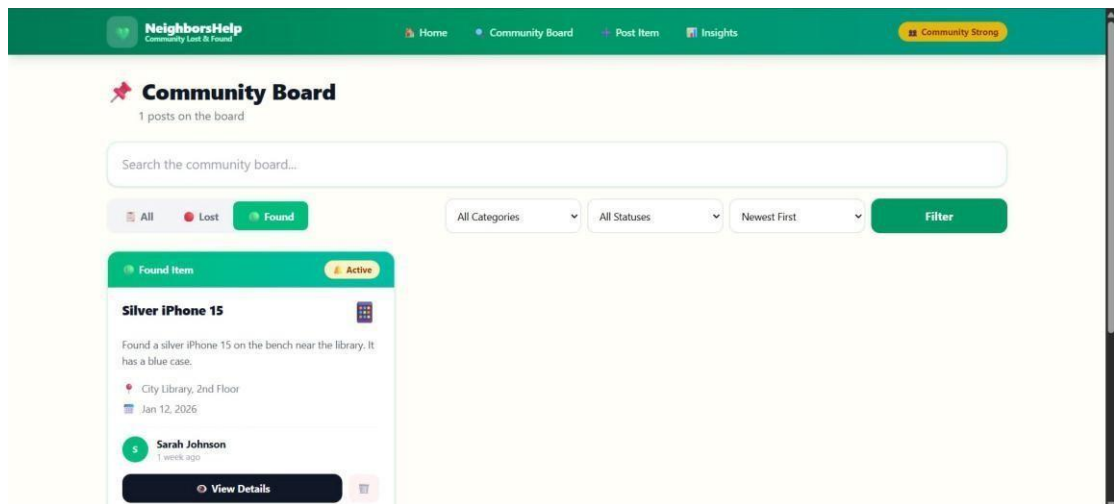
Lost & Found System Architecture Diagram



Interface Design/Screenshot







Database Structure

List of Tables

1. Users Table

Store user account information

Field Name	Data Type	Description
id	INT	Primary key
name	VARCHAR	Full name of the user
email	VARCHAR	User email address
password	VARCHAR	Encrypted password
role	VARCHAR	Admin/user
created_at	TIMESTAMP	Date created
updated_at	TIMESTAMP	Date updated

2.ItemTable

Store lost and found item

Field Name	Data Type	Description
item_id	INT	Primary key
user_id	INT	Owner/Reporter
Item_name	VARCHAR	Full name of patient
description	TEXT	Item details
category	VARCHAR	Gender
status	VARCHAR	Lost or found
location	TEXT	Where item was lost/found
date_reported	DATE	Date submitted

3.ReportTable

Store user report

Field Name	Data Type	Description
Report_id	INT	Primary key
item_id	INT	Linked item
user_id	INT	Report
Report_type	VARCHAR	Lost or Found
details	TEXT	Report details
create_at	TIMESTAMP	Date created

4.ClaimsTable

Store Claim requests

Field Name	Data Type	Description
claim_id	INT	Primary key
item_id	INT	Linked patient ID
User_id	TEXT	Patient diagnosis
status	TEXT	Pending/approved/rejected
Date_claimed	DATE	Claim date

Implementation and Testing

Implementation

The Lost & Found System was developed using Laravel, PHP, and MySQL. The frontend was built using HTML, CSS, JavaScript, and Bootstrap to ensure responsiveness and usability.

The system allows users to register, log in, and submit reports for lost or found items. Administrators manage item records and verify claims. All data is stored securely in the database.

Testing

Functional Testing

- User login and registration tested
- Item reporting tested
- Search functionality tested
- Admin management features tested

Database Testing

- Verified correct data storage
- Checked table relationships
- Ensured data consistency

User Interface Testing

- Tested responsiveness
- Verified form validation
- Checked navigation links

Problems Encountered

- Database connection issues
- Login authentication errors
- Minor bugs in search functionality

Solutions Applied

- Fixed configuration settings
- Improved validation logic
- Debugged backend functions
- Retested system until stable

Conclusion

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The Lost & Found System was successfully developed to improve the management of lost and found items. It provides a centralized, efficient, and organized platform for reporting and tracking items.

The system reduces manual processes, improves accuracy, and makes it easier for users to find and claim lost belongings. It also enhances communication between users and administrators.

Summary

The system integrates reporting, searching, and claiming of lost and found items into one platform. It improves efficiency, reduces errors, and enhances item tracking.

What Was Achieved in the Project

The project successfully developed a Lost & Found System that allows users to report and manage lost items digitally. It simplifies the process of item tracking and improves overall organization.

Benefits of the System

- Faster item tracking
- Reduced manual paperwork
- Better organization of records
- Easier item claiming process
- Improved communication

Learning Gained During Development

The development of this system improved skills in web development, database management, and system design using Laravel, PHP, and MySQL. It also strengthened problem-solving and analytical thinking skills. **Recommendations**

Future Improvements

- Mobile application version
- Email/SMS notifications
- QR code system for item tagging

- AI-based item matching
- Advanced search filters

Additional Features

Image upload for items

- User chat system
- Automated claim verification
- Dashboard analytics

System Enhancement Ideas

- Cloud-based deployment
- Multi-location support
- Improved security features
- Real-time updates